

CSC15S Assignment 9 – Security Design Principles III and LDAP

Total: 100pts

What to turn in: zip and upload all your related files to **Assignment 9** in Blackboard. Your zip should include the following files:

- (1) CSC15Slab9.docx (with all questions completed)
- (2) log_Mission7.txt

Part 1. Challenges of Cybersecurity in Varieties of Computing Environment

[20 pts. Estimated time to complete: 10 minutes.]

1. Give a brief example of the cybersecurity challenge in varieties of computing environment. [10 pts]

Securing cloud environments is a major challenge due to risks like data breaches from misconfigured storage or unauthorized access. The diversity of platforms makes consistent security difficult.

2. Briefly describe the existing partnership of federal, state, local and industry for cyber defense. [10 pts]

Programs like CISA and MS-ISAC enable collaboration between governments and private sectors for threat intelligence sharing, critical infrastructure protection, and rapid incident response.

Part 2. Dr. Evil's Training Room. Mission 7 – Godfather Part II

[80 pts. Estimated time to complete: 1.5 hrs]

Michael managed to move the family to Vegas. The used-to-be gang-empire has been rebuilt as a modern enterprise in computing. Michael decides to provide quality computing service to all its members. Everybody will be able to sign on into any computer in the network. Return your favor to Michael and help him set up SSO.

Please note that all of your config will be applied to the Vegas image (169.254.251.1), but the EvTrainer should be run on the main image. Once you feel ready to submit, please attach log_Mission7.txt to Blackboard. The following are some additional hints for each of the challenges.

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Mission 7: Godfather Part II (80 pts)
Name: s
Grade: 80.0 pts
Time Spent: 427 seconds

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QID  Title                Pts    Time  QID  Title                Pts    Time
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1    Vegas Headquarter    15.0/15  1      2    Corleone Family      30.0/30  3
3    SSO                   35.0/35  423

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Enter question ID (e.g., 1) to redo a question or hit Q to quit
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Challenge	Hint
1. Vegas Headquarter	Q1 (Warm-up): make sure the network and DNS are working; Q2 (LDAP Server): simply follow the instructions in the given tutorial; Q3 (Reconfigure): follow the instructions on when to use the DNS format (e.g., corleone.org) and when to use LDAP format (e.g., dc=corleone,dc=ogr)
2. Corleone Family	Q1 (No miners): you'll need to chop off some of the entries in the input file; Q2 (Idscripts): Make sure to follow the instructions exactly, especially do not use extra spaces in your strings. Q3 (Inhuman VIM Editor): the trick is that VIM inserts an additional char into the file, which causes the authentication problem for Idascripts to access the LDAP database. Q4 (artists and philanthropists): simple! Need to create groups first.
3. SSO	Make sure to take a snapshot of the Vegas Image before the question! Q1 (Name Switch Service): please follow the instructions given exactly (even if some of the instruction are given in purpose to trick you); Q2 (Diagnosis): you will excise some interesting DEBUGGING efforts, e.g., using strace to analyze network activities. The key is to use the RIGHT TYPE of socket and protocol when accessing the remote LDAP server; Q3 (SSO): Your last step is to

	fix the server address (use the ldap protocol instead) and make the LDAP based authentication work.
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